

Maths 3rd SEM, (MA2003-1) Question Bank

Unit 1

1. A particle moves along a curve $x = t^3$, $y = 1 + t^2$ and $z = 2t$ where t is the time variable. Find the components of velocity and acceleration at $t = 2$ in the direction $2\hat{i} + \hat{j} + \hat{k}$
2. Prove that the cylindrical co-ordinate system is an orthogonal curvilinear coordinate system.
3. Prove that the spherical co-ordinate system is an orthogonal curvilinear coordinate system.
4. State and prove Greens theorem in a plane.
5. Find the unit normal vector to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$
6. A particle moves along the curve $x = e^t$, $y = 2\cos 3t$, $z = 2\sin 3t$ where t is the time variable. Find the magnitude of velocity and acceleration at $t = 0$
7. Using Stoke's theorem evaluate $\oint_C (x + y)dx + (2x - z)dy + (y + z)dz$, where C is the boundary of a triangle with vertices $(2,0,0)$, $(0,3,0)$ and $(0,0,0)$.
8. If $\phi = x^3 + y^3 + z^3 - 3xyz$ and $\vec{F} = \nabla\phi$. Find divergence of $\vec{F}$ and curl of $\vec{F}$
9. Prove that $\text{div}\left(\frac{\vec{r}}{r^3}\right) = 0$
10. Obtain the directional derivative of the function $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ along the vector $2\hat{i} - \hat{j} - 2\hat{k}$
11. A particle moves along a curve $x = 7 + 8t^5$, $y = 4t^2$ and $z = 5 + t$, where t is the time variable. Find the components of velocity and acceleration at $t = 2$ in the direction $\hat{i} + 3\hat{j} + \hat{k}$
12. Determine the constants a , b , c so that the vector function $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational.
13. Find the work done in moving a particle once round the circle C in the xy -plane with the centre at the origin and radius 3, in the force field $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y)\hat{k}$
14. If $\vec{F} = (5xy - 6x^2)\hat{i} + (2y - 4x)\hat{j}$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where C is the curve $y = x^3$ in the xy plane from the point $(1, 1)$ to $(2, 8)$.
15. Define a conservative vector field. Show that the vector field $\vec{F} = (y + z)\hat{i} + (x + z)\hat{j} + (x + y)\hat{k}$ is a conservative vector field. Find the scalar potential.

16. Show that the vector field $\vec{F} = (2xyz^2)\hat{i} + (x^2z^2 + z\cos yz)\hat{j} + (2x^2yz + y\sin yz)\hat{k}$ is conservative. Find its scalar potential.
17. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, find ∇r^n
18. If $\phi = x^3 + y^3 + z^3 + yz$ and $\vec{F} = \nabla\phi$, find $\Delta \bullet \vec{F}$ and $\Delta \times \vec{F}$
19. Using Gauss divergence theorem evaluate $\iint_C \vec{F} \bullet \vec{n} \, ds$ where $\vec{F} = (x)\hat{i} + (y^2)\hat{j} + (z^2)\hat{k}$, S is the surface bounded by $x^2 + y^2 = 4, z = 0$ and $z = 3$.
20. Find the value of 'a' if the vector field $\vec{F} = (ax^2y + yz)\hat{i} + (xy^2 - xz^2)\hat{j} + (2xy - 2x^2y^2)\hat{k}$ has zero divergence.
21. Find $\nabla\phi$ and $|\nabla\phi|$ for $\phi = 2x^3y^2z^4$ at (3,-1,2)
22. Using Gauss divergence theorem evaluate $\iint_S \vec{F} \bullet \vec{n} \, ds$, where $\vec{F} = (y - z + 2)\hat{i} - (yz + 4)\hat{j} - (zx)\hat{k}$, and S is the closed surface of the cube $x=0, x=2, y=0, y=2, z=0, z=2$.
23. Using Stokes theorem evaluate $\oint_C \vec{F} \cdot d\vec{r}$ for $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ over a rectangle C in the xy plane bounded by $x=0, x=a, y=0$, and $y=b$
24. If $\phi = 2x^3y^2z^4$ and $\vec{F} = \nabla\phi$, find $\Delta \bullet \vec{F}$ and $\Delta \times \vec{F}$

Unit 2

1. Find the bilinear transformation which maps the points $z = 1, i, -1$ onto $w = i, 0, -i$
2. Find the bilinear transform which maps the points $z = 0, 1, i$ onto $w = 1+i2, -1, 2-i$
3. If $f(z)$ is analytic, then prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4|f'(z)|^2$
4. Show that the $f(z) = e^z$ is analytic and find $f'(z)$
5. Show that $f(z) = \cos(z)$ is an analytic function and find $f'(z)$
6. Determine the analytic function $f(z) = u + iv$ with an imaginary part $v = e^{-x}(x \cos y + y \sin y)$
7. If $f(z) = u(x, y) + iv(x, y)$ is analytic, show that its real and imaginary parts are harmonic.
8. Determine the analytic function $f(z) = u + iv$ with an imaginary part $v = \frac{x-y}{x^2-y^2}$
9. If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then prove that at any point of D, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
10. Show that $u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2$ is harmonic.

11. Define harmonic function, check whether the function $u = x^3 - 3xy^2 + 3x^2 - 3y^2$ is harmonic.
12. Evaluate $\oint_C \frac{e^z}{(z+1)^2} dz$ using Cauchy integral formula, where C is a circle given by $|z| = 2$
13. State and prove Cauchy Riemann equations in cartesian form.
14. State and prove Cauchy's integral formula.
15. Using Cauchy integral formula evaluate $\int_C \frac{zdz}{(z-5)(2z+3)}$ where C is the circle $|z| = 4$
16. State Cauchy residue theorem and use it to evaluate $\int_C \frac{(z^2)dz}{(z-1^2)(z+2)}$ where c is the circle $|z| = \frac{5}{2}$
17. Using Cauchy integral formula evaluate $\int_C \frac{(3z^2+2)dz}{(z-1)(z^2+9)}$ where C is the circle $|z - 1| = 2$
18. Evaluate $\oint_C \frac{(2z^2+1)}{(z+1)^2(z-2)} dz$ in the region $|z| = 3$
19. Find the Laurent's series expansion of $f(z) = \frac{z+1}{(z+2)(z+3)}$ in the region i) $|z| < 2$ and ii) $2 < |z| < 3$
20. Write the Laurent's series which represent $f(z) = \frac{z+3}{z(z+1)(z-2)}$ in (i) $|z| < 1$ (ii) $1 < |z| < 2$
21. Find the Laurent's expansion of $f(z) = \frac{z^2-1}{(z^2+5z+6)}$ in (i) $|z| < 2$ (ii) $|z| > 3$
22. Find the poles and residues of $f(z) = \frac{z^2}{(z^2+3z+2)^2}$
23. Obtain the poles and residues of $f(z) = \frac{2z}{z^2-z-2}$

Unit 3- (Fourier Transforms)

1. Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$
2. Find the Fourier cosine transform of $f(x) = e^{-x}$
3. Obtain the Fourier sine transform of $f(x) = e^{-|x|}$
4. Obtain the Fourier cosine transform of e^{-x^2}
5. Find the Fourier transform of $f(x) = e^{-a^2x^3}$, $a > 0$
6. State and prove convolution theorem for Fourier Transform.